Sharper Upper Bounds for the Easier Waring Problem via Weighted Fifth-Power Identities, Non-Ideal Prouhet–Tarry–Escott Multigrades, and Local Obstruction Bypass

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November 3, 2026

Abstract

The Easier Waring Problem, formulated by E. M. Wright in 1934, asks for each integer $k \geq 2$ to determine the minimum number of terms $s = v(k)$ such that every integer $n \in \mathbb{Z}$ can be represented as an algebraic sum of s signed k-th powers: $n = \epsilon_1 x_1^k + \epsilon_2 x_2^k + \cdots + \epsilon_s x_s^k$ with $x_i \in \mathbb{Z}_{\geq 0}$ and $\epsilon_i \in \{+1, -1\}$. Following a comprehensive literature audit and arithmetic verification of modular polynomial identity methods, we resolve several foundational theoretical gaps and establish sharper upper bounds across small and moderate exponents. First, for $k = 5$, we incorporate the weighted linear polynomial identity method to bypass the classical Kleiman–Wright divisibility barrier $720 \mid C$ that traps unit-weight Prouhet–Tarry–Escott (PTE) identities at $v(5) \leq 10$; using an 8-summand (7-distinct-form) $(2, 2, 1, 1, 1, 1, 1, 1)$-weighted identity with square-free slope $C_{\mathrm{I}} = 3,844,230 = 2 \cdot 3 \cdot 5 \cdot 13 \cdot 9857$ (and a parametric companion with $C_{\mathrm{II}} = 7,327,885,530$) satisfying $\Delta(5, C) = 1$, we prove the unconditional bound $v(5) \leq 9$, breaking an 87-year barrier. Second, we prove that the Chinese Remainder Theorem decomposition $\Delta(k, C) = \max_{p^l \| C} \Delta(k, p^l)$ holds unconditionally for odd k but fails in general for even k due to cross-modular sign-split incompatibility, and we establish the exact Signed-CRT Decomposition Theorem for even k along with the general $p^a \parallel k$ Hensel Stabilization Lemma. Third, for $k = 11$, we prove via the Rees–Smyth Multiplicity Lemma computation of Coppersmith, Mossinghoff, Scheinerman, and VanderKam (2024) ($23 \mid C_{10}(\mathbb{Z})$) that the Sophie Germain prime $p = 23$ divides C for every ideal degree-9 PTE solution over $\mathbb{Z}$ (locking ideal-PTE bounds at $v(11) \leq 31$), and we bypass this obstruction via Wróblewski's non-ideal size-12 multigrade 8.12.36 (providing a corrected proof of Wróblewski's parametric family) to establish $\Delta(11, C) = \Delta(11, 11^2) = 4$ and $v(11) \leq 28$, plus a conditional parity-sign reduction to $v(11) \leq 27$. Fourth, for $k = 14$, we bypass the symmetric ideal-PTE obstruction $29 \mid C'_{12}(\mathbb{Z})$ ($v(14) \leq 54$) via the non-ideal-derived multigrade 11.20.107 to establish $v(14) \leq 45$, and we complete the verified set of explicit Frolov bounds across $9 \leq k \leq 17$: $v(9) \leq 29$, $v(10) \leq 30$, $v(11) \leq 28$, $v(12) \leq 33$, $v(14) \leq 45$, $v(15) \leq 57$, and $v(17) \leq 64$ (repaired via the authentic catalogue entry 14.30.153). Complete reproducible certification programs for all even-k exhaustive signed-CRT computations are provided.

1 Introduction and Literature Review

Let $k \geq 2$ be a positive integer. In 1934, E. M. Wright [25] formulated the *Easier Waring Problem*: to determine the least positive integer $s = v(k)$ such that every integer $n \in \mathbb{Z}$ admits a representation of the form

$$n = \epsilon_1 x_1^k + \epsilon_2 x_2^k + \cdots + \epsilon_s x_s^k, \quad x_i \in \mathbb{Z}_{\geq 0}, \quad \epsilon_i \in \{+1, -1\}. \tag{1}$$

When k is odd, $(-x)^k = -x^k$, so (1) is equivalent to representing every $n \in \mathbb{Z}$ as a sum of s k-th powers of signed integers $x_i \in \mathbb{Z}$. Whereas the classical Waring constants $g(k)$ and $G(k)$

restrict all signs ϵ_i to $+1$, permitting negative signs fundamentally alters both the local arithmetic obstructions and the applicable analytic and algebraic methods [12, 2, 24]. In the classical Waring problem, the non-negativity constraint restricts candidate representations to an archimedean box $0 \le x_i \le n^{1/k}$, enabling finite exhaustion and circle-method major-arc lattice counting. Allowing summands of opposite signs destroys this compact box constraint, producing non-compact hyperboloids of representations. Consequently, $v(2) = 3$ remains the only exponent for which $v(k)$ is known exactly [25, 24].

1.1 The Frolov–Wright Algebraic Paradigm and State of the Art

Two broad methodologies have been applied to bound $v(k)$ across exponents: analytic circle-method reductions and algebraic polynomial identities. Analytically, Wright [25] observed that if $G(k)$ denotes the classical threshold above which every integer is a sum of non-negative k-th powers, then writing $n = (y^k + n) - y^k$ yields $v(k) \le G(k) + 1$. Although modern Vinogradov–Wooley smooth-number circle-method bounds [22] show that $v(k) = O(k \log k)$ asymptotically, for moderate exponents $3 \le k \le 17$ the resulting numerical bounds—such as $v(5) \le 18$, $v(11) \le 68$, and $v(14) \le 93$ [23]—are far weaker than those derived from algebraic polynomial identities.

The dominant algebraic mechanism for bounding $v(k)$ stems from the Prouhet–Tarry–Escott (PTE) problem [10, 11, 26, 2]. Two integer multisets $A = \{a_1, \dots, a_s\}$ and $B = \{b_1, \dots, b_s\}$ form a PTE solution of size s and degree m, written $A \overset{m}{=} B$, if their power-sum differences vanish up to degree m:

$$S_r(A, B) := \sum_{i=1}^{s} a_i^r - \sum_{i=1}^{s} b_i^r = 0 \quad \text{for all } 1 \le r \le m.$$

When $A \overset{k-2}{=} B$, General M. M. Frolov [10, 11] discovered that the higher-order difference polynomial

$$P(x) = \sum_{i=1}^{s} (x + a_i)^k - \sum_{i=1}^{s} (x + b_i)^k = \sum_{r=0}^{k} \binom{k}{r} S_r(A, B) x^{k-r} = Cx + D \tag{2}$$

collapses to a linear polynomial in x with non-zero slope $C = kS_{k-1}(A, B)$ and constant term $D = S_k(A, B)$ [26, 2, 24]. When the solution possesses central symmetry ($A = -A, B = -B$ or $A = -B$), the system reduces to half-size even-power or odd-power multigrades (*mehrgradige Gleichungen*) [10, 27]. By reducing modulo $|C|$, Wright [26] formulated the modular Frolov bound: if $\Delta(k, |C|)$ denotes the local signed Waring constant modulo $|C|$ (the minimal number of signed k-th powers required to represent every residue class modulo $|C|$), then

$$v(k) \le 2s + \Delta(k, |C|) \quad (C = kS_{k-1}(A, B) \ne 0). \tag{3}$$

In his 1965 survey, Wright [26] tabulated upper bounds including $v(9) \le 34$, $v(10) \le 38$, $v(12) \le 48$, $v(13) \le 50$, $v(15) \le 82$, and $v(17) \le 94$. In the wider literature on the Easier Waring Problem, several small-exponent bounds have long been established outside the PTE framework. Notably:

- $k = 4$: Hunter (1941) [30] established $9 \le v(4) \le 10$, improving upon earlier estimates.
- $k = 6$: K. Subba Rao (1934) [29] proved $6 \le v(6) \le 14$ via an explicit sixth-power algebraic identity, a result later systematized by modern expositions.
- Survey of small-k bounds: Vavilov (2022) [24] provides a comprehensive modern survey recording these small-k ranges, detailing the Rao sixth-power identity and Vaserstein's eighth-power constructions.

- $k = 8$: Vaserstein (1988) [21] proved that every integer is a sum or difference of 28 eighth powers ($v(8) \leq 28$), which Habsieger (1991) [14] conceptualized via the sign representation of S_3 acting on (u, v, w). In the same year, Ajai Choudhry (1991) [31] independently published an alternative 12-term eighth-power polynomial identity proving the identical 28-term bound $v(8) \leq 28$.

Turning specifically to Frolov/PTE polynomial identity improvements in the moderate exponent range $k \geq 5$, progress over the past several decades was confined to isolated cases:

- $k = 5$ ($v(5) \leq 10$, Fuchs and Wright 1939 [12]): An 8-term monic degree-3 PTE identity with slope $C = 720$ and $\Delta(5, 720) = 2$ established $v(5) \leq 10$ [15], an upper bound that remained unimproved in the literature for 87 years.
- $k = 13$ ($v(13) \leq 27$, Choudhry and Wróblewski 2008 [7]): Using the smallest known ideal degree-11 symmetric PTE solution 10.12.151 of size $s = 12$ ($2s = 24$) discovered by Kuosa, Meyrignac, and Chen Shuwen in 1999, Choudhry and Wróblewski proved that the resulting slope C_{13} satisfies $\Delta(13, C_{13}) = \Delta(13, 13^3) = 3$, giving $v(13) \leq 24 + 3 = 27$.

1.2 Literature Audit: Arithmetic Obstructions and Theoretical Gaps

A comprehensive audit of the literature on modular polynomial identities reveals three foundational obstructions and theoretical gaps that have constrained the field:

1. The Local-to-Global Signed-CRT Gap for Even Exponents. In prior literature (including [24, §4.3]), it was asserted without qualification that the local signed Waring constant decomposes across coprime prime powers as $\Delta(k, C) = \max_{p^l \| C} \Delta(k, p^l)$. While this multiplicative decomposition holds unconditionally for odd exponents k (where signs can be absorbed into the base via $\epsilon_i x_i^k = (\epsilon_i x_i)^k$), it fails in general for even exponents k. When k is even, $(-x)^k = +x^k$ cannot absorb signs, so signed representations across distinct coprime prime powers $p_j^{l_j}$ require mutually compatible partitions of signs into positive and negative counts. Disjoint sign-split requirements across coprime prime powers lead to strict inflation $\Delta(k, C) > \max_j \Delta(k, p_j^{l_j})$, as demonstrated by the corrected counterexample $k = 4, C = 65$ (Remark 2.4) where $\Delta(4, 5) = \Delta(4, 13) = 2$ but $\Delta(4, 65) = 3$. A rigorous Signed-CRT theory and Hensel stabilization lemma are therefore required.

2. The Kleiman–Wright Divisibility Barrier at $k = 5$. For $k = 5$, any monic (unit-weight) degree-3, size-4 PTE identity $P(z) = \sum_{i=1}^4 (z + \alpha_i)^5 - \sum_{i=1}^4 (z + \beta_i)^5 = Cz + D$ has slope $C = 5S_4(\alpha, \beta) = -20C_4(\alpha, \beta)$. By Bastien's classical theorem and Kleiman's theorem [28, 18], $36 = 2^2 \cdot 3^2 \mid C_4(\alpha, \beta)$, forcing $720 = 2^4 \cdot 3^2 \cdot 5 \mid C$ and $360 \mid D$ for every ideal degree-3 PTE solution over $\mathbb{Z}$ [12, 15]. Because $9 \mid C$ and $16 \mid C$, 3-adic and 2-adic obstructions force $\Delta(5, C) \geq \Delta(5, 9) = 2$, permanently trapping all classical unit-weight identities at $v(5) \leq 8 + 2 = 10$. Bypassing this barrier requires abandoning unit weights and constructing weighted linear identities whose slope C is square-free and coprime to all primes $p \equiv 1 \pmod 5$.

3. Inescapable Sophie Germain Divisibility Barriers in Ideal PTE Solutions ($k = 11, 14$). By Bastien's theorem [2, 17], any degree-m PTE solution satisfies $s \geq m + 1$; solutions with $s = m + 1$ are called ideal. For an ideal solution (A, B) of size n and degree $n - 1$, Newton's symmetric polynomial identities imply that $\prod_{i=1}^n (x - a_i) - \prod_{i=1}^n (x - b_i) = (-1)^n (e_n(A) - e_n(B)) =: C_n(A, B)$ is a constant, and the first non-vanishing power-sum difference is [18, 5]

$$S_n(A, B) = (-1)^{n-1} n C_n(A, B), \quad C = (n+1) S_n(A, B) = (-1)^{n-1} n(n+1) C_n(A, B). \tag{4}$$

Rees and Smyth [18] introduced the Multiplicity Lemma over $\mathbb{F}_p[x]$ to prove that additional primes must divide $C_n(\mathbb{Z}) := \gcd\{C_n(A,B) : (A,B) \text{ ideal of size } n \text{ over } \mathbb{Z}\}$. Recently, Coppersmith, Mossinghoff, Scheinerman, and VanderKam [9, Section 2, Table 1] proved via exhaustive evaluation over degree-12 polynomials in $\mathbb{F}_{23}[x]$ that the Sophie Germain prime $p = 23 = 2 \cdot 11 + 1$ divides $C_{10}(\mathbb{Z})$ for every ideal degree-9 ($n = 10$) PTE solution over $\mathbb{Z}$. Consequently, $23 \mid C$ is an unavoidable structural invariant, forcing $\Delta(11, |C|) \geq \Delta(11, 23) = 11$ and locking ideal-PTE bounds at $v(11) \leq 20 + 11 = 31$. Similarly, for $k = 14$, Coppersmith et al. confirmed that $29 \mid C'_{12}(\mathbb{Z})$ for every symmetric ideal degree-11 PTE solution over $\mathbb{Z}$, forcing standard Tarry-shifted size-20 solutions to inherit $29 \mid C_{14}$ and locking bounds at $v(14) \leq 40 + 14 = 54$. Bypassing these barriers requires stepping outside the ideal regime into non-ideal multigrades where $C_n(x)$ is non-constant.

1.3 Summary of Audit Corrections and New Contributions

In this paper, we resolve these arithmetic obstructions and theoretical gaps, establishing the following results:

1. **Exact Signed-CRT Decomposition for Even k and $p^a \parallel k$ Hensel Stabilization (Section 2):** We prove that while $\Delta(k, C) = \max_{p^l \parallel C} \Delta(k, p^l)$ holds unconditionally for odd k, it fails in general for even k due to sign-split incompatibility across coprime prime powers. We establish the exact Signed-CRT Criterion (Theorem 2.3) in terms of intersecting admissible sign-split sets $S_r(k, p_j^{l_j}; u)$ and prove the complete Hensel Stabilization Lemma (Lemma 2.5) covering both $p \nmid k$ and $p^a \parallel k$ ($a \geq 1$), clarifying the necessary unit-summand hypothesis. We also provide the corrected modulus-65 counterexample (Remark 2.4).

2. **Unconditional Proof of $v(5) \leq 9$ via Weighted Linear Fifth-Power Identities (Section 3):** Developing the weighted polynomial method (initiated in the author's preliminary preprint [13]), we establish why ideal monic degree-3 size-4 PTE identities are trapped at $v(5) \leq 10$, prove the universal $30 \mid C$ and $15 \mid D$ divisibility theorem and the $\mathbb{F}_{11}$-Bastien rank-defect theorem, and supply full proofs for the 8-summand $(2, 2, 1, 1, 1, 1, 1, 1)$-weighted identity with $C_{\mathrm{I}} = 3,844,230 = 2 \cdot 3 \cdot 5 \cdot 13 \cdot 9857$ satisfying $\Delta(5, C) = 1$, unconditionally establishing $v(5) \leq 9$.

3. **Inescapable Modulo-23 Ideal-PTE Barrier and Non-Ideal Bypass $v(11) \leq 28$ (Section 4):** We prove that $23 \mid C_{10}(\mathbb{Z})$ [9] forces $\Delta(11, |C|) \geq 11$ and $v(11) \leq 31$ for every ideal degree-9 PTE solution over $\mathbb{Z}$. Stepping to non-ideal size $s = 12$ ($2s = 24$) via Wróblewski's 8.12.36 [27], we give a verified algebraic proof of Wróblewski's parametric system, evaluate $\Delta(11, 11^2) = 4$ via balanced ternary expansions, and establish $v(11) \leq 28$. We also analyze the conditional parity-sign reduction to $v(11) \leq 27$, correct the Newton-identity relation for D, and prove the exact translation-invariance condition.

4. **Modulo-29 Bypass for $k = 14$ ($v(14) \leq 45$) and Unified Frolov Bounds for $9 \leq k \leq 17$ (Sections 5–6):** We prove how Tarry shifting an ideal degree-11 solution inherits $29 \mid C'_{12}(\mathbb{Z})$ [9], locking bounds at $v(14) \leq 54$. Using the non-ideal-derived multigrade 11.20.107, we bypass $p = 29$, prove the $r = 4$ local obstruction, and rigorously establish the global signed-CRT constant $\Delta(14, C_{14}) = 5$ via full prime-power reduction and exhaustive bitmask certification, proving $v(14) \leq 45$. We also provide corrected, verified proofs of $v(9) \leq 29$, $v(10) \leq 30$, $v(12) \leq 33$, $v(15) \leq 57$, and $v(17) \leq 64$ (repaired via the authentic multigrade 14.30.153 and local generator $3^{17} \equiv 224 \pmod{289}$).

5. **Reproducibility and Exhaustive CRT Verification Suite (Section 7):** We provide an auditable computational certificate and self-contained algorithm for the even-k exhaustive bitmask CRT exhaustions ($k = 10, 12, 14$) and catalogue moment verifications.

2 General Symmetric Reconstruction, Signed-CRT Theory, and Hensel Stabilization

To ensure complete rigor across both odd and even exponents k, we first state the general multigrade reconstruction lemma (allowing unequal half-set sizes $m_1 + m_2 = s$, as needed for asymmetric odd-power multigrades in [27]) and establish the exact local-to-global modular theory for $\Delta(k, C)$.

Lemma 2.1 (General Symmetric Multigrade Reconstruction; cf. [10, 27, 24])**.** *Let $h \geq 1$ be an integer.*

(i) ***(Even Multigrade to Odd Degree** $2h+1$**):** *If disjoint multisets $A_+ = \{a_1, \ldots, a_m\}$ and $B_+ = \{b_1, \ldots, b_m\}$ of positive integers satisfy $\sum_{i=1}^m a_i^{2r} = \sum_{i=1}^m b_i^{2r}$ for all $1 \leq r \leq h$, then $A = A_+ \cup (-A_+)$ and $B = B_+ \cup (-B_+)$ satisfy $A \overset{2h+1}{=} B$ of size $s = 2m$, with $S_{2h+3}(A, B) = 0$ and $S_{2h+2}(A, B) = 2\sum_{i=1}^m (a_i^{2h+2} - b_i^{2h+2})$.*

(ii) ***(Odd Multigrade to Even Degree** $2h$ **with General Split** $m_1 + m_2 = s$**):** *Suppose disjoint multisets $A_{\text{odd}} = \{a_1, \ldots, a_{m_1}\}$ and $B_{\text{odd}} = \{b_1, \ldots, b_{m_2}\}$ satisfy $A_{\text{odd}} \cap (-A_{\text{odd}}) = \emptyset$, $B_{\text{odd}} \cap (-B_{\text{odd}}) = \emptyset$, and $\sum_{i=1}^{m_1} a_i^{2r-1} = \sum_{j=1}^{m_2} b_j^{2r-1}$ for all $1 \leq r \leq h$. Then the multisets*

$$A = A_{\text{odd}} \cup (-B_{\text{odd}}), \quad B = B_{\text{odd}} \cup (-A_{\text{odd}}) = -A$$

are disjoint of equal size $s = m_1 + m_2$ and satisfy $A \overset{2h}{=} B$, with $S_{2h+2}(A, B) = 0$ and

$$S_{2h+1}(A, B) = 2\left[\sum_{i=1}^{m_1} a_i^{2h+1} - \sum_{j=1}^{m_2} b_j^{2h+1}\right].$$

Moreover, for any shift $t \in \mathbb{Z}$ and scale divisor $g = \gcd(\{a - t : a \in A\} \cup \{b - t : b \in B\}) \geq 1$, the affine-reduced integer multisets $\tilde{A} = \{(a - t)/g : a \in A\}$ and $\tilde{B} = \{(b - t)/g : b \in B\}$ satisfy $\tilde{A} \overset{2h}{=} \tilde{B}$ with $S_{2h+1}(\tilde{A}, \tilde{B}) = g^{-(2h+1)} S_{2h+1}(A, B)$ and $S_{2h+2}(\tilde{A}, \tilde{B}) = -(2h + 2)(t/g) S_{2h+1}(\tilde{A}, \tilde{B})$.

Proof. Part (i) is immediate since odd power sums vanish identically on $A = -A$ and $B = -B$. For Part (ii), both $A = A_{\text{odd}} \cup (-B_{\text{odd}})$ and $B = -A$ have cardinality $s = m_1 + m_2$. For any even exponent $2r \geq 2$, $(-x)^{2r} = x^{2r}$ implies $\sum_{x \in A} x^{2r} = \sum_{a \in A_{\text{odd}}} a^{2r} + \sum_{b \in B_{\text{odd}}} b^{2r} = \sum_{y \in B} y^{2r}$, while for each odd exponent $2r - 1$ ($1 \leq r \leq h$), $\sum_{x \in A} x^{2r-1} = \sum_{a \in A_{\text{odd}}} a^{2r-1} - \sum_{b \in B_{\text{odd}}} b^{2r-1} = 0 = \sum_{y \in B} y^{2r-1}$. Finally, expanding $((x - t)/g)^j = g^{-j} \sum_{r=0}^j \binom{j}{r} (-t)^{j-r} x^r$ and using $S_r(A, B) = 0$ for $0 \leq r \leq 2h$ along with $S_{2h+2}(A, -A) = 0$ yields the exact relations for $S_{2h+1}(\tilde{A}, \tilde{B})$ and $S_{2h+2}(\tilde{A}, \tilde{B})$. □

Definition 2.2 (Local Signed Waring Constant and Admissible Sign-Splits)**.** For integers $k \geq 1$ and $m \geq 1$, let $\Delta(k, m)$ (denoted $v(k, m)$ in [24]) be the least non-negative integer r such that every residue class $u \in \mathbb{Z}/m\mathbb{Z}$ can be written as

$$u \equiv \epsilon_1 y_1^k + \epsilon_2 y_2^k + \cdots + \epsilon_r y_r^k \pmod{m}, \quad y_i \in \mathbb{Z}, \quad \epsilon_i \in \{+1, -1\}. \tag{5}$$

For even k, given $r \geq 0$ and $u \in \mathbb{Z}/m\mathbb{Z}$, define the admissible positive-sign count set

$$S_r(k, m; u) := \left\{ P \in \{0, 1, \ldots, r\} : u \equiv \sum_{i=1}^{P} y_i^k - \sum_{j=1}^{r-P} z_j^k \pmod{m} \text{ for some } y_i, z_j \in \mathbb{Z} \right\}.$$

(Note that since $0 = \pm 0^k$, if $P_0 \in S_{r_0}(k, m; u)$ for some $r_0 \leq r$, then $P_0 + a \in S_r(k, m; u)$ for every $0 \leq a \leq r - r_0$.)

Theorem 2.3 (Exact Signed-CRT Decomposition of $\Delta(k, C)$ for Odd and Even Exponents)**.** *Let $C = p_1^{l_1} p_2^{l_2} \dots p_q^{l_q} > 1$ be the prime-power factorization of C.*

(i) ***(Odd Exponents** k**):** If k is odd, then*

$$\Delta(k, C) = \max_{1 \le j \le q} \Delta(k, p_j^{l_j}). \tag{6}$$

(ii) ***(Even Exponents** k**):** If k is even, then a residue class $u \in \mathbb{Z}/C\mathbb{Z}$ is a sum of r signed k-th powers modulo C if and only if*

$$\bigcap_{j=1}^{q} S_r(k, p_j^{l_j}; u \bmod p_j^{l_j}) \neq \emptyset. \tag{7}$$

Consequently, $\Delta(k, C)$ is the least integer $r \ge \max_{1 \le j \le q} \Delta(k, p_j^{l_j})$ such that (7) holds for every tuple $(u_1, \dots, u_q) \in \prod_{j=1}^{q} (\mathbb{Z}/p_j^{l_j}\mathbb{Z})$. In particular, for even k, strict inequality $\Delta(k, C) > \max_{1 \le j \le q} \Delta(k, p_j^{l_j})$ occurs whenever two prime powers $p_a^{l_a}, p_b^{l_b} \parallel C$ possess residues u_a, u_b with disjoint admissible sign-split sets at $r = \max_j \Delta(k, p_j^{l_j})$.

Proof. (i) Let $M = \max_{1 \le j \le q} \Delta(k, p_j^{l_j})$. Reducing any mod-$C$ representation modulo $p_j^{l_j}$ gives $\Delta(k, C) \ge M$. Conversely, when k is odd, $(-y)^k = -y^k$, so any representation $u \equiv \sum_{i=1}^{M} \epsilon_{j,i} y_{j,i}^k \pmod{p_j^{l_j}}$ can be written with all positive signs as $u \equiv \sum_{i=1}^{M} (\epsilon_{j,i} y_{j,i})^k \pmod{p_j^{l_j}}$. By the Chinese Remainder Theorem (CRT), choosing $Y_i \in \mathbb{Z}$ with $Y_i \equiv \epsilon_{j,i} y_{j,i} \pmod{p_j^{l_j}}$ for each $1 \le i \le M$ yields $u \equiv \sum_{i=1}^{M} Y_i^k \pmod{C}$, proving $\Delta(k, C) = M$.

(ii) Suppose k is even. If $u \equiv \sum_{i=1}^{r} \epsilon_i Y_i^k \pmod{C}$ with $P = |\{i : \epsilon_i = +1\}|$ and $r - P = |\{i : \epsilon_i = -1\}|$, reducing modulo each $p_j^{l_j}$ shows that $P \in S_r(k, p_j^{l_j}; u \bmod p_j^{l_j})$ for every $j = 1, \dots, q$, so the intersection (7) contains P and is non-empty. Conversely, if there exists $P \in \bigcap_{j=1}^{q} S_r(k, p_j^{l_j}; u \bmod p_j^{l_j})$, then for each j there exist integers $y_{j,1}, \dots, y_{j,P}$ and $z_{j,1}, \dots, z_{j,r-P}$ such that $u \equiv \sum_{i=1}^{P} y_{j,i}^k - \sum_{m=1}^{r-P} z_{j,m}^k \pmod{p_j^{l_j}}$. By CRT, there exist integers $Y_1, \dots, Y_P$ and $Z_1, \dots, Z_{r-P}$ with $Y_i \equiv y_{j,i} \pmod{p_j^{l_j}}$ and $Z_m \equiv z_{j,m} \pmod{p_j^{l_j}}$ for all j, yielding $u \equiv \sum_{i=1}^{P} Y_i^k - \sum_{m=1}^{r-P} Z_m^k \pmod{C}$. □

Remark 2.4 (Explicit Counterexample to Naive CRT for Even k)**.** For $k = 4$ and $C = 5 \cdot 13 = 65$, the local constants are $\Delta(4, 5) = 2$ (with fourth powers $y^4 \in \{0, 1\} \pmod 5$) and $\Delta(4, 13) = 2$ (with fourth powers $y^4 \in \{0, 1, 3, 9\} \pmod{13}$). Consider the residue class $u = 7 \pmod{65}$, which corresponds to $u \equiv 2 \pmod 5$ and $u \equiv 7 \pmod{13}$. At length $r = 2$:

- Modulo 5: $u \equiv 2 \pmod 5$ can only be reached as $1^4 + 1^4 \equiv 2$, forcing both signs to be positive. Thus $S_2(4, 5; 2) = \{2\}$.
- Modulo 13: $u \equiv 7 \pmod{13}$. The sum of two fourth powers with $P = 2$ yields residues in $\{0, 1, 2, 3, 4, 5, 6, 9, 10, 12\}$, which does not contain 7. However, with $P = 1$ we have $2^4 - 3^4 \equiv 3 - 9 = -6 \equiv 7 \pmod{13}$, and with $P = 0$ we have $-2^4 - 2^4 \equiv -3 - 3 = -6 \equiv 7 \pmod{13}$. Thus $S_2(4, 13; 7) = \{0, 1\}$.

Consequently, $S_2(4, 5; 2) \cap S_2(4, 13; 7) = \{2\} \cap \{0, 1\} = \emptyset$, so $u = 7$ cannot be represented by two signed fourth powers modulo 65. Exhaustive enumeration of two signed fourth powers modulo 65 misses exactly the 6 residue classes $\{7, 18, 28, 37, 47, 58\}$, whereas three signed fourth powers cover all 65 classes. Therefore $\Delta(4, 65) = 3 > \max\{\Delta(4, 5), \Delta(4, 13)\} = 2$.

Next, we state and prove the complete Hensel Stabilization Lemma covering both $p \nmid k$ and $p^a \parallel k$.

Lemma 2.5 (Complete Hensel Lifting and Stabilization for $\Delta(k, p^l)$)**.** *Let $k \geq 2$ be an integer, p a prime, and $l \geq 1$.*

(i) ***(Single-Term Mod-p Case for $p > 2, p \nmid k$):*** *Suppose $p > 2$, $p \nmid k$, and $\Delta(k, p) = 1$ (which holds whenever $\gcd(k, p-1) = 1$, or when k is even, $\gcd(k, p-1) = 2$, and $p \equiv 3 \pmod 4$ so $-1 \notin (\mathbb{F}_p^\times)^2$). Then every unit in $(\mathbb{Z}/p^l\mathbb{Z})^\times$ is a single signed k-th power, and $\Delta(k, p^l) = 2$ for all $l \geq 2$.*

(ii) ***(General Coprime Case $p > 2, p \nmid k$):*** *Suppose $p > 2$, $p \nmid k$, and $\Delta(k, p) \geq 2$. Then $\Delta(k, p^l) = \Delta(k, p)$ for all $l \geq 1$. In particular, if $p = 2k + 1$ is prime (k odd or even), then $y^k \in \{0, \pm 1\} \pmod p$ for all $y \in \mathbb{Z}$, and $\Delta(k, p^l) = \Delta(k, p) = \frac{p-1}{2} = k$ for all $l \geq 1$.*

(iii) ***(2-adic Moduli):*** *If k is odd, then $\Delta(k, 2) = 1$ and $\Delta(k, 2^l) = 2$ for all $l \geq 2$. If k is even with $2^a \parallel k$ ($a \geq 1$), then for every odd integer y, $y^k \equiv 1 \pmod{2^{a+2}}$, and the set of odd k-th powers modulo 2^l stabilizes for all $l \geq a + 2$ as the subgroup $\{u \in (\mathbb{Z}/2^l\mathbb{Z})^\times : u \equiv 1 \pmod{2^{a+2}}\}$.*

(iv) ***(Ramified Case $p^a \parallel k$ with $p > 2, a \geq 1$):*** *Suppose $p > 2$ and $p^a \parallel k$ with $a \geq 1$. Then for any $y \in \mathbb{Z}$ with $p \nmid y$ and any $e \geq a + 1$,*

$$(y + p^{e-a}t)^k \equiv y^k + p^e(k/p^a)y^{k-1}t \pmod{p^{e+1}}$$

lifts every unit k-th power modulo p^{a+1} surjectively to all its lifts modulo p^l for all $l \geq a + 1$. Consequently, if $\Delta(k, p^{a+1}) \geq 2$ and every residue $u \in \mathbb{Z}/p^{a+1}\mathbb{Z}$ admits a representation of length $r = \Delta(k, p^{a+1})$ containing at least one unit summand $\epsilon_1 y_1^k$ with $p \nmid y_1$, then $\Delta(k, p^l) = \Delta(k, p^{a+1})$ for all $l \geq a + 1$.

Proof. (i) Since $\Delta(k, p) = 1$, for every unit $u \not\equiv 0 \pmod p$ there exist $\epsilon \in \{+1, -1\}$ and $y_0 \not\equiv 0 \pmod p$ such that $f(y_0) := \epsilon y_0^k - u \equiv 0 \pmod p$. Since $p \nmid k$ and $p \nmid y_0$, $f'(y_0) = \epsilon k y_0^{k-1} \not\equiv 0 \pmod p$, so Hensel's Lemma lifts y_0 uniquely to $y \in (\mathbb{Z}/p^l\mathbb{Z})^\times$ with $\epsilon y^k \equiv u \pmod{p^l}$. For $l \geq 2$, $u = p \bmod p^2$ is not a single signed k-th power (since $p \mid y \implies p^k \mid y^k \implies p^2 \mid y^k$), whereas for any multiple $u \equiv 0 \pmod p$, $u - 1 \not\equiv 0 \pmod p$ is a unit and thus $u = (u-1) + 1^k$ is a sum of 2 signed k-th powers modulo p^l.

(ii) Let $r = \Delta(k, p) \geq 2$. Every $u \not\equiv 0 \pmod p$ has a mod-p representation of length r with some $y_1 \not\equiv 0 \pmod p$, while for $u \equiv 0 \pmod p$ we take $1^k - 1^k + 0^k + \cdots + 0^k \equiv 0 \pmod p$ with $y_1 = 1 \not\equiv 0 \pmod p$. Since $p \nmid k$, $g(y_1) = \epsilon_1 y_1^k + \sum_{i=2}^r \epsilon_i y_i^k - u$ satisfies $g'(y_1) = \epsilon_1 k y_1^{k-1} \not\equiv 0 \pmod p$, so Hensel's Lemma lifts y_1 to $\mathbb{Z}/p^l\mathbb{Z}$ for all $l \geq 1$. When $p = 2k + 1$ is prime, Fermat's Little Theorem gives $(y^k)^2 = y^{p-1} \equiv 1 \pmod p$ for $p \nmid y$, so $y^k \in \{0, \pm 1\} \pmod p$ and r signed k-th powers cover $[-r, r] \bmod p$, requiring $2r + 1 \geq p = 2k + 1 \iff r \geq k$.

(iii) When k is odd, $\gcd(k, 2^{l-1}) = 1$ implies $y \mapsto y^k$ is an automorphism of $(\mathbb{Z}/2^l\mathbb{Z})^\times$, so every odd residue mod 2^l is a k-th power and every even residue $2m = (2m-1) + 1^k$ is a sum of two odd k-th powers, while 2 mod 2^l ($l \geq 2$) is not a single k-th power. When $2^a \parallel k$ ($a \geq 1$), Euler's theorem and binomial induction give $(1 + 2^e t)^k \equiv 1 + 2^{e+a}(k/2^a)t \pmod{2^{e+a+1}}$ for all $e \geq 2$, which lifts every residue $\equiv 1 \pmod{2^{a+2}}$ to all its lifts modulo 2^l for $l \geq a + 2$.

(iv) Write $k = p^a k_0$ with $p \nmid k_0$. For $e \geq a + 1$ and $r \geq 2$, the p-adic valuation of $\binom{k}{r} p^{r(e-a)}$ is $a - v_p(r!) + r(e-a) \geq a + r(e-a) - \frac{r-1}{p-1} \geq e + 1$ since $e - a \geq 1$ and $p \geq 3$. Thus $(y + p^{e-a}t)^k \equiv y^k + p^e k_0 y^{k-1} t \pmod{p^{e+1}}$. Because $p \nmid k_0 y_1^{k-1}$, varying $t \in \mathbb{Z}/p\mathbb{Z}$ hits all p lifts of $y_1^k \bmod p^e$ to $\mathbb{Z}/p^{e+1}\mathbb{Z}$. Induction on $e \geq a + 1$ lifts any representation $u \equiv \epsilon_1 y_1^k + \sum_{i=2}^r \epsilon_i y_i^k \pmod{p^{a+1}}$ with $p \nmid y_1$ to $\mathbb{Z}/p^l\mathbb{Z}$ for all $l \geq a + 1$. □

3 Unconditional Proof of $v(5) \leq 9$ via Weighted Fifth-Power Identities

We now incorporate the weighted linear polynomial identity method (developed in [13]) to prove $v(5) \leq 9$, improving upon the classical bound $v(5) \leq 10$ of Fuchs and Wright [12].

3.1 Why Classical Unit-Weight PTE Identities Are Trapped at $v(5) \leq 10$

In a classical monic (unit-weight) degree-3, size-4 PTE identity $P(z) = \sum_{i=1}^{4}(z+\alpha_i)^5 - \sum_{i=1}^{4}(z+\beta_i)^5 = Cz + D$ (where $\sum_{i=1}^{4}\alpha_i^r = \sum_{i=1}^{4}\beta_i^r$ for $r = 1,2,3$), equation (4) gives $C = 5S_4(\alpha,\beta) = -20C_4(\alpha,\beta)$. By Kleiman's theorem [28] (and direct modulo-16 and modulo-9 residue analysis [12]), $36 = 2^2 \cdot 3^2 \mid C_4(\alpha,\beta)$, so $720 = 2^4 \cdot 3^2 \cdot 5 \mid C$ and $360 \mid D$ hold for every ideal degree-3 PTE solution over $\mathbb{Z}$ (achieved minimally by $(z-4)^5 + (z-1)^5 + z^5 + (z+3)^5 - 2(z-3)^5 - 2(z+2)^5 = 720z - 360$ [15, Theorem 407]). Since $3^2 \mid C$ and $2^4 \mid C$, Lemma 2.5(i),(iii) forces $\Delta(5,C) \geq \Delta(5,9) = 2$, permanently locking any unit-weight PTE bound at $v(5) \leq 8+2 = 10$.

To achieve $\Delta(5,C) = 1$, we must instead construct an 8-summand weighted linear identity

$$\sum_{i=1}^{m} s_i(A_iz + B_i)^5 = Cz + D, \quad A_i \in \mathbb{Z}_{>0}, \quad B_i \in \mathbb{Z}, \quad s_i \in \mathbb{Z} \setminus \{0\}, \tag{8}$$

of total weight $\sum_{i=1}^{m}|s_i| = 8$ whose slope C is square-free and has no prime divisor $p \equiv 1 \pmod 5$. We first prove that $30 \mid C$ universally and that $\sum_{i=1}^{m}|s_i| = 8$ (hence $s = 8+1 = 9$) is the exact minimum for any such univariate linear identity.

Proposition 3.1 (Universal $30 \mid C$ Divisibility and $\mathbb{F}_{11}$-Bastien Rank-Defect Obstruction)**.** *Let $\sum_{i=1}^{m} s_i(A_iz + B_i)^5 = Cz + D$ ($C \neq 0$) be any linear polynomial identity in $\mathbb{Z}[z]$ with $\sum_{i=1}^{m}|s_i| = M$.*

(i) ***(Universal Slope Divisibility):*** *$30 \mid C$ holds for every $m \geq 1$, and for each $p \in \{2,3,5\}$, $D \equiv \sum_{p \mid A_i} s_iB_i \pmod p$. In particular, if $\gcd(A_i,15) = 1$ for all i, then $15 \mid \gcd(C,D)$.*

(ii) ***($\mathbb{F}_{11}$-Bastien Obstruction for $M \leq 7$):*** *If $M \leq 7$, then $11 \mid C$, forcing $\Delta(5,|C|) \geq \Delta(5,11) = 5$ and $M + \Delta(5,|C|) \geq 9$. Thus $M = 8$ (giving $v(5) \leq 8+1 = 9$) is the absolute minimum summand count for a univariate linear identity with $11 \nmid C$.*

Proof. (i) Vanishing of $[z^3]$ gives $10\sum_{i=1}^{m} s_iA_i^3B_i^2 = 0 \implies \sum_{i=1}^{m} s_iA_i^3B_i^2 = 0$, while $C = 5\sum_{i=1}^{m} s_iA_iB_i^4$. Since $AB^2(B^2-A^2) = AB^2(B-A)(B+A)$ is always divisible by 6, we have

$$\frac{C}{5} = \sum_{i=1}^{m} s_iA_iB_i^2(B_i^2 - A_i^2) + \sum_{i=1}^{m} s_iA_i^3B_i^2 = \sum_{i=1}^{m} s_iA_iB_i^2(B_i^2 - A_i^2) \equiv 0 \pmod 6 \implies 30 \mid C.$$

Similarly, vanishing of $[z^4]$ gives $\sum_{i=1}^{m} s_iA_i^4B_i = 0$; subtracting from $D = \sum_{i=1}^{m} s_iB_i^5$ modulo $p \in \{2,3,5\}$ (where $B_i^5 \equiv B_i \pmod p$ and $A_i^4 \equiv 1 \pmod p$ for $p \nmid A_i$) yields $D \equiv \sum_{p \mid A_i} s_iB_i \pmod p$.

(ii) Suppose $M = \sum_{i=1}^{m}|s_i| \leq 7$ (repeating terms according to $|s_i|$ so each $s_i \in \{+1,-1\}$). Modulo 11, any term with $11 \mid A_i$ drops out of $[z^5], [z^4], [z^3], [z^2], [z^1] = C$. For the remaining $M' \leq 7$ terms with $11 \nmid A_i$, since $y^5 \in \{\pm 1\} \pmod{11}$, absorbing the sign of A_i^5 mod 11 into s_i and setting $\gamma_i \equiv B_iA_i^{-1} \in \mathbb{F}_{11}$ reduces $[z^5] \equiv 0 \pmod{11}$ to $\sum_{i=1}^{M'} s_i \equiv 0 \pmod{11}$. Since $|s_i| = 1$ and $M' \leq 7 < 11$, we must have $\sum_{i=1}^{M'} s_i = 0$ in $\mathbb{Z}$, so $M' = 2k$ is even with $k \leq 3$ positive signs S_+ and $k \leq 3$ negative signs S_-. Vanishing of $[z^4], [z^3], [z^2]$ modulo 11 then gives $\sum_{i \in S_+} \gamma_i^r \equiv \sum_{j \in S_-} \gamma_j^r \pmod{11}$ for $r = 1,2,3$. Since $k \leq 3 < 11$, Newton's identities over $\mathbb{F}_{11}$ force the multisets $\{\gamma_i : i \in S_+\}$ and $\{\gamma_j : j \in S_-\}$ to be identical in $\mathbb{F}_{11}$, whence $C \equiv 5\left[\sum_{i \in S_+} \gamma_i^4 - \sum_{j \in S_-} \gamma_j^4\right] \equiv 0 \pmod{11}$. Since $y^5 \in \{0, \pm 1\} \pmod{11}$, $\Delta(5,11) = 5$. □

3.2 Direct Integer Gap-Sum Formulation and Proof of $v(5) \leq 9$

To eliminate the $2^4 \cdot 3^2$ factor from C at $M = 8$, we weight the first pair of fifth powers by $A_1 = A_2 = 2$ and keep the remaining three pairs at unit weight $A_3 = \cdots = A_8 = 1$:

$$P_8(z) = (2z + B_1)^5 - (2z + B_2)^5 + \sum_{i=2}^{4} \left[(z + B_{2i-1})^5 - (z + B_{2i})^5\right], \quad B_j \in \mathbb{Z}. \tag{9}$$

Proposition 3.2 (Integer Gap-Sum System and Root-Free Parameterization)**.** *Set $\delta_i = B_{2i-1} - B_{2i} \in \mathbb{Z}$ and $\sigma_i = B_{2i-1} + B_{2i} \in \mathbb{Z}$ with $\sigma_i \equiv \delta_i \pmod 2$ for $i = 1, 2, 3, 4$.*

(i) The polynomial $P_8(z)$ in (9) is linear in z ($[z^5] = [z^4] = [z^3] = [z^2] = 0$) if and only if

$$16\delta_1 + \delta_2 + \delta_3 + \delta_4 = 0, \tag{10}$$

$$8\delta_1\sigma_1 + \delta_2\sigma_2 + \delta_3\sigma_3 + \delta_4\sigma_4 = 0, \tag{11}$$

$$12\delta_1\sigma_1^2 + 3(\delta_2\sigma_2^2 + \delta_3\sigma_3^2 + \delta_4\sigma_4^2) = -(4\delta_1^3 + \delta_2^3 + \delta_3^3 + \delta_4^3). \tag{12}$$

(ii) Choosing the parametric gap vector $(\delta_1, \delta_2, \delta_3, \delta_4) = (n, 4n^2, -16n, -4n^2)$ and sum vector

$$(\sigma_1, \sigma_2, \sigma_3, \sigma_4) = \left(-\frac{n^2 - 4n + 341}{2}, 2n, \frac{n^2 + 4n - 341}{4}, 0\right)$$

satisfies (10)–(12) identically over $\mathbb{Q}[n]$ and yields for every odd integer n an integral 8-term identity $P_8(z; n) = C(n)z + D(n)$ with slope

$$C(n) = \frac{30n^3(85n^4 - 678n^2 + 116281)}{8} \equiv 2 \pmod 4 \quad (\forall n \equiv 1 \pmod 2). \tag{13}$$

Proof. (i) Writing $B_{2i-1} = (\sigma_i + \delta_i)/2$ and $B_{2i} = (\sigma_i - \delta_i)/2$, direct binomial expansion of $(A_i z + B_{2i-1})^5 - (A_i z + B_{2i})^5$ gives $[z^5] = 2^5 - 2^5 + 1 - 1 + 1 - 1 + 1 - 1 = 0$, $[z^4] = 5(16\delta_1 + \delta_2 + \delta_3 + \delta_4)$, $[z^3] = 5(8\delta_1\sigma_1 + \delta_2\sigma_2 + \delta_3\sigma_3 + \delta_4\sigma_4)$, and $[z^2] = \frac{5}{2}[12\delta_1\sigma_1^2 + 3(\delta_2\sigma_2^2 + \delta_3\sigma_3^2 + \delta_4\sigma_4^2) + (4\delta_1^3 + \delta_2^3 + \delta_3^3 + \delta_4^3)]$.

(ii) For the stated vectors (δ_i) and (σ_i), we check:

- $16\delta_1 + \delta_2 + \delta_3 + \delta_4 = 16n + 4n^2 - 16n - 4n^2 = 0$;
- $8\delta_1\sigma_1 + \delta_2\sigma_2 + \delta_3\sigma_3 + \delta_4\sigma_4 = -4n(n^2 - 4n + 341) + 8n^3 - 4n(n^2 + 4n - 341) + 0 = 0$;
- Using $(n^2 - 4n + 341)^2 - (n^2 + 4n - 341)^2 = (2n^2)(-8n + 682) = -16n^3 + 1364n^2$, the left-hand side of (12) equals $3n[(n^2 - 4n + 341)^2 - (n^2 + 4n - 341)^2] + 48n^4 = 4092n^3 = -(4n^3 + 64n^6 - 4096n^3 - 64n^6) = -(4\delta_1^3 + \delta_2^3 + \delta_3^3 + \delta_4^3)$.

When n is odd, $n^2 \equiv 1 \pmod 8$, so each $B_{2i-1}, B_{2i} = (\sigma_i \pm \delta_i)/2$ is an integer, and $85n^4 - 678n^2 + 116281 \equiv 85 - 678 + 116281 = 115688 \equiv 8 \pmod{16}$, ensuring $2^1 \parallel C(n)$. □

Lemma 3.3 (Bijectivity Criterion $\Delta(5, C) = 1$)**.** *For any integer $C > 1$, $\Delta(5, C) = 1$ (every residue modulo C is a single fifth power $y^5 \bmod C$) if and only if C is square-free and every prime factor $p \mid C$ satisfies $p \not\equiv 1 \pmod 5$. (For the trivial modulus $C = 1$, there is only the zero class and $\Delta(5, 1) = 0$.)*

Proof. By Theorem 2.3(i), $\Delta(5, C) = \max_{p^e \| C} \Delta(5, p^e)$. If $e \geq 2$, then $p \bmod p^e$ is not a fifth power ($p \mid y^5 \implies p \mid y \implies p^5 \mid y^5$), so $\Delta(5, p^e) \geq 2$. If $e = 1$, $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic of order $p - 1$, so $y \mapsto y^5$ is a bijection on $\mathbb{Z}/p\mathbb{Z}$ if and only if $\gcd(5, p - 1) = 1 \iff p \not\equiv 1 \pmod 5$ (including $p = 5$, where $y^5 \equiv y \pmod 5$ by Fermat's Little Theorem). □

Theorem 3.4 (Unconditional Upper Bound $v(5) \leq 9$)**.** *Every integer $N \in \mathbb{Z}$ can be expressed as an algebraic sum of at most 9 fifth powers of non-negative integers:*

$$5 \leq v(5) \leq 9. \tag{14}$$

Specifically:

(i) ***(Compact 8-Summand Identity with*** $\max|B_i| = 39$***):*** *The gap-sum solution $(\delta_1, \delta_2, \delta_3, \delta_4) = (-1, 4, 28, -16)$ and $(\sigma_1, \sigma_2, \sigma_3, \sigma_4) = (-75, 74, -16, 28)$ of Proposition 3.2(i) yields the 8-summand (7-distinct-form) polynomial identity*

$$\begin{aligned}(2z-38)^5 - (2z-37)^5 + (z+39)^5 - (z+35)^5 \\ - (z-22)^5 - (z+22)^5 + 2(z+6)^5 = C_{\mathrm{I}} z + D_{\mathrm{I}},\end{aligned} \tag{15}$$

where

$$C_{\mathrm{I}} = 3,844,230 = 2 \cdot 3 \cdot 5 \cdot 13 \cdot 9857, \quad D_{\mathrm{I}} = 27,826,665.$$

(ii) ***(Parametric 8-Summand Identity):*** *The 8-term identity*

$$\begin{aligned}(2z-4167)^5 - (2z-4174)^5 + (z+9)^5 - (z+5)^5 + (z-2141)^5 \\ - (z-2029)^5 + (z-2)^5 - (z+2)^5 = C_{\mathrm{II}} z + D_{\mathrm{II}}\end{aligned} \tag{16}$$

holds identically with

$$C_{\mathrm{II}} = 7,327,885,530 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 17 \cdot 2052629, \quad D_{\mathrm{II}} = -10,178,299,260,675.$$

(iii) Both C_{I} and C_{II} satisfy $\Delta(5, C_{\mathrm{I}}) = \Delta(5, C_{\mathrm{II}}) = 1$. Hence for every $N \in \mathbb{Z}$, there exists a unique integer $y \in (-C_{\mathrm{I}}/2, C_{\mathrm{I}}/2]$ such that $y^5 \equiv N - D_{\mathrm{I}} \pmod{C_{\mathrm{I}}}$, and setting $z = (N - D_{\mathrm{I}} - y^5)/C_{\mathrm{I}} \in \mathbb{Z}$ represents $N = (C_{\mathrm{I}} z + D_{\mathrm{I}}) + y^5$ as an algebraic sum of $8 + 1 = 9$ fifth powers.

Proof. Expanding the left-hand side of (15) by binomial coefficients $\binom{5}{r}$ verifies each degree in z:

- $[z^5]$: $(32 - 32) + (1 - 1 - 1 - 1 + 2) = 0$;
- $[z^4]$: $5[16(-38) - 16(-37) + 39 - 35 - (-22) - 22 + 2(6)] = 5(-16 + 4 + 22 - 22 + 12) = 0$;
- $[z^3]$: $10[8(38^2 - 37^2) + (39^2 - 35^2) - (22^2 + 22^2) + 2(6^2)] = 10(600 + 296 - 968 + 72) = 0$;
- $[z^2]$: $10[4((-38)^3 - (-37)^3) + (39^3 - 35^3) - ((-22)^3 + 22^3) + 2(6^3)] = 10(-16876 + 16444 + 432) = 0$;
- $[z^1]$: $5[2(38^4 - 37^4) + (39^4 - 35^4) - 2(22^4) + 2(6^4)] = 5(421950 + 812816 - 468512 + 2592) = 3,844,230 = C_{\mathrm{I}}$;
- $[z^0]$: $(-38)^5 - (-37)^5 + 39^5 - 35^5 + 2(6^5)$

$$\begin{aligned}&= -79,235,168 + 69,343,957 + 90,224,199 - 52,521,875 + 15,552 \\ &= 27,826,665 = D_{\mathrm{I}}.\end{aligned}$$

Direct expansion identically confirms (16). Both C_{I} and C_{II} are square-free products of primes. For C_{I}, the prime factors $\{2, 3, 5, 13, 9857\} \equiv \{2, 3, 0, 3, 2\} \pmod 5$ contain no prime $\equiv 1 \pmod 5$. For C_{II}, $\{2, 3, 5, 7, 17, 2052629\} \equiv \{2, 3, 0, 2, 2, 4\} \pmod 5$ likewise avoids primes $\equiv 1 \pmod 5$. Lemma 3.3 yields $\Delta(5, C_{\mathrm{I}}) = \Delta(5, C_{\mathrm{II}}) = 1$, proving $v(5) \leq 8 + 1 = 9$. □

4 The Inescapable Modulo-23 Ideal-PTE Obstruction at $k = 11$ and Non-Ideal Bypass Proving $v(11) \leq 28$

We now turn to $k = 11$ and prove first that applying Wright's modular Frolov method (3) to any ideal degree-9 PTE solution ($s = 10$) over $\mathbb{Z}$ cannot improve upon $v(11) \leq 31$, and second that stepping to non-ideal size $s = 12$ bypasses this prime obstruction to achieve $v(11) \leq 28$.

4.1 The Ideal-PTE Barrier $v(11) \leq 31$

Proposition 4.1 (The Ideal-PTE Barrier $v(11) \leq 31$)**.** *For every ideal PTE solution (A, B) of size $s = 10$ and degree 9 over $\mathbb{Z}$ (whether symmetric or non-symmetric), the Frolov linear coefficient $C = 11S_{10}(A, B) = -110C_{10}(A, B)$ is divisible by the Sophie Germain prime $p = 23 = 2 \cdot 11 + 1$, and consequently*

$$2s + \Delta(11, |C|) \geq 20 + \Delta(11, 23) = 20 + 11 = 31. \tag{17}$$

Moreover, the Borwein–Lisoňek–Percival ideal symmetric solution 8.10.313 [4, 27], defined by $A_+ = \{99, 100, 188, 301, 313\}$ and $B_+ = \{71, 131, 180, 307, 308\}$, achieves this bound and gives $v(11) \leq 31$.

Proof. By the Rees–Smyth Multiplicity Lemma [9, Lemma 2.1], if $p > n$ is a prime not dividing $C_n(A, B) = \prod_{i=1}^{n}(x - a_i) - \prod_{i=1}^{n}(x - b_i)$ for some ideal PTE solution (A, B) of size n over $\mathbb{Z}$, then there exists a polynomial $h \in \mathbb{F}_p[x]$ of degree exactly $p - n - 1$ such that $h(a) \equiv \nu_a(q_1) - \nu_a(q_2) \pmod{p}$ for all $a \in \mathbb{F}_p$ (where $q_1(x) = \prod_{i=1}^{n}(x - a_i)$ and $q_2(x) = \prod_{i=1}^{n}(x - b_i)$), and therefore

$$\sum_{a=0}^{p-1} |\langle h(a) \rangle_p| \leq 2n, \tag{18}$$

where $\langle m \rangle_p \in (-p/2, p/2]$ denotes the centered residue modulo p. For $n = 10$ and $p = 23$, the required degree is $p - n - 1 = 12$. By exhaustive verification over all degree-12 polynomials $h \in \mathbb{F}_{23}[x]$, Coppersmith, Mossinghoff, Scheinerman, and VanderKam [9, Section 2.1, Table 1] proved that the minimum of $\sum_{a=0}^{22} |\langle h(a) \rangle_{23}|$ over all degree-12 polynomials in $\mathbb{F}_{23}[x]$ is 24. Since $2n = 20 < 24$, inequality (18) cannot hold for $n = 10$, which proves that $23 \mid C_{10}(\mathbb{Z})$, i.e., $23 \mid C_{10}(A, B)$ for every ideal PTE solution (A, B) of size 10 over $\mathbb{Z}$.

By Newton's identity (4) at $n = 10$ [18, 5], $C = 11S_{10}(A, B) = -110C_{10}(A, B)$, so $23 \mid C$. Since $23 = 2 \cdot 11 + 1$ is prime, Lemma 2.5(ii) gives $\Delta(11, 23) = 11$, and Theorem 2.3(i) yields $\Delta(11, |C|) \geq \Delta(11, 23) = 11$, establishing (17).

Finally, for the ideal symmetric solution 8.10.313 with $A_+ = \{99, 100, 188, 301, 313\}$ and $B_+ = \{71, 131, 180, 307, 308\}$, direct computation gives

$$\begin{aligned} |C_{\text{ideal}}| &= 22 \sum_{i=1}^{5} (a_i^{10} - b_i^{10}) = 625,599,997,396,881,002,496,000 \\ &= 2^{15} \cdot 3^6 \cdot 5^3 \cdot 7^2 \cdot 11^3 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 37. \end{aligned}$$

Here $\Delta(11, 23) = 11$, $\Delta(11, 11^3) = 4$ (by Lemma 4.3 and Lemma 2.5(iv) below), and $\Delta(11, p^l) \leq 2$ for all other prime power factors $p^l \parallel C_{\text{ideal}}$ since $\gcd(11, p - 1) = 1$. Thus $\Delta(11, |C_{\text{ideal}}|) = 11$ and (3) gives $v(11) \leq 20 + 11 = 31$. □

4.2 Corrected Proof of Wróblewski's Parametric Family and $v(11) \leq 28$

Once we step from ideal size $s = 10$ to non-ideal size $s = 12$ for degree 9, the polynomial difference $\prod_{i=1}^{12}(x - a_i) - \prod_{i=1}^{12}(x - b_i)$ has degree $12 - 10 = 2$ rather than 0, so the ideal

constant obstruction $23 \mid C_{10}(\mathbb{Z})$ no longer applies. We first give a complete, corrected proof of the algebraic validity of Wróblewski's parametric multigrade family [27, Page 12] that generates our size-12 solution 8.12.36.

Proposition 4.2 (Wróblewski's Parametric Family for Degree-9, Size-12 Multigrades [27])**.** *Let $a, b, c, m, n \in \mathbb{Z}$ satisfy the two Diophantine equations*

$$2(a^2+b^2+c^2) = m^2+11n^2, \tag{19}$$

$$(a+b+c)(a+b-c)(a-b+c)(-a+b+c) = 8n^2(n^2-m^2) = -8n^2(m^2-n^2). \tag{20}$$

Then the two 6-element multisets

$$L = \{2a, 2b, 2c, 4n, |m-n|, m+n\},$$
$$R = \{|a+b+c|, |a-b+c|, |a+b-c|, |a-b-c|, |m-3n|, m+3n\}$$

satisfy $\sum_{x\in L} x^{2r} = \sum_{y\in R} y^{2r}$ for $2r = 2, 4, 6, 8$. In particular, choosing $(a, b, c, m, n) = (30, 31, 7, 16, 18)$ satisfies (19)–(20) and yields, after dividing all entries of L and R by 2, the solution 8.12.36:

$$A_+ = \{1, 7, 17, 30, 31, 36\}, \quad B_+ = \{3, 4, 19, 27, 34, 35\}. \tag{21}$$

Proof. Let $L_1 = \{2a, 2b, 2c\}$, $L_2 = \{4n, m-n, m+n\}$, $R_1 = \{a+b+c, a-b+c, a+b-c, a-b-c\}$, and $R_2 = \{m-3n, m+3n\}$. For $2r = 2$,

$$\sum_{x\in L_1} x^2 = 4(a^2+b^2+c^2) = \sum_{y\in R_1} y^2,$$
$$\sum_{x\in L_2} x^2 = 16n^2 + (m-n)^2 + (m+n)^2 = 2m^2+18n^2 = (m-3n)^2 + (m+3n)^2 = \sum_{y\in R_2} y^2,$$

so $\sum_{x\in L} x^2 = \sum_{y\in R} y^2$ holds identically for all a, b, c, m, n.

For $2r = 4, 6, 8$, let $S = a^2+b^2+c^2$, $Q = a^4+b^4+c^4$, $P_6 = a^6+b^6+c^6$, $P_8 = a^8+b^8+c^8$, and $e_3 = a^2b^2c^2$, and set

$$\begin{aligned} H &= (a+b+c)(a+b-c)(a-b+c)(-a+b+c) \\ &= 2(a^2b^2+b^2c^2+c^2a^2) - (a^4+b^4+c^4) = S^2 - 2Q. \end{aligned}$$

Notice first that in R_1, expanding $\sum_{y\in R_1} y^{2r}$ over the four sign choices $(+1, \pm 1, \pm 1)$ kills all odd exponents of a, b, c: By Newton's identities for the three variables (a^2, b^2, c^2) with elementary symmetric polynomials $e_1 = S$, $e_2 = \frac{1}{4}(S^2+H)$, and $e_3 = a^2b^2c^2$, we have

$$Q = \frac{1}{2}(S^2-H), \quad P_6 = \frac{1}{4}S(S^2-3H) + 3e_3, \quad P_8 = \frac{1}{8}(S^4 - 4S^2H + H^2) + 4Se_3.$$

Simultaneously, in R_1, the four squares y^2 have elementary symmetric polynomials in y^2 expressible purely in (S, H, e_3), and remarkably the $e_3 = a^2b^2c^2$ terms in $\sum_{x\in L_1} x^{2r} = 2^{2r}P_{2r}$ and $\sum_{y\in R_1} y^{2r}$ match identically for both $2r = 6$ (where $2^6(3e_3) = 192e_3 = \sum_{y\in R_1} y^6|_{e_3}$) and $2r = 8$ (where $2^8(4Se_3) = 1024Se_3 = \sum_{y\in R_1} y^8|_{e_3}$)! Explicitly, direct expansion in $\mathbb{Z}[S, H, e_3, m, n]$ gives the exact polynomial identities:

$$\sum_{x\in L} x^4 - \sum_{y\in R} y^4 = -12[H + 8n^2(m^2-n^2)],$$
$$\sum_{x\in L} x^6 - \sum_{y\in R} y^6 = -60S[H + 8n^2(m^2-n^2)] - 240n^2(m^2-n^2)(2S - m^2 - 11n^2),$$
$$\begin{aligned} \sum_{x\in L} x^8 - \sum_{y\in R} y^8 &= 28(H - 8S^2)[H + 8n^2(m^2-n^2)] \\ &\quad - 56n^2(m^2-n^2)(2S + m^2 + 11n^2)(2S - m^2 - 11n^2). \end{aligned}$$

Under conditions (19) ($2S - m^2 - 11n^2 = 0$) and (20) ($H + 8n^2(m^2 - n^2) = 0$), all three differences vanish identically.

At $(a, b, c, m, n) = (30, 31, 7, 16, 18)$, we verify (19) and (20):

$$2(30^2 + 31^2 + 7^2) = 2(900 + 961 + 49) = 3820 = 16^2 + 11(18^2),$$
$$68 \cdot 54 \cdot 6 \cdot 8 = 176,256 = 8(18^2)(18^2 - 16^2).$$

Dividing $L = \{60, 62, 14, 72, 2, 34\}$ and $R = \{68, 6, 54, 8, 38, 70\}$ by 2 yields A_+ and B_+ in (21). □

Lemma 4.3 (Exact Evaluation of $\Delta(11, 11^l)$ via Balanced Ternary Arithmetic)**.** *For $k = 11$ and every $l \geq 2$, the local signed Waring constant modulo 11^l is*

$$\Delta(11, 11^l) = \Delta(11, 121) = 4. \tag{22}$$

Specifically, the non-zero signed 11-th powers modulo $121 = 11^2$ form the 10-element multiplicative subgroup of signed powers of 3:

$$H_{11} = \{\pm 1, \pm 3, \pm 9, \pm 27, \pm 81\} \equiv \{\pm 1, \pm 3, \pm 9, \pm 27, \pm 40\} \pmod{121}. \tag{23}$$

Exactly 111 residue classes in $\mathbb{Z}/121\mathbb{Z}$ are sums of at most 3 elements of H_{11}, and the remaining 10 residue classes—namely $\mathcal{E}_{121} = \{\pm 20, \pm 47, \pm 56, \pm 59, \pm 60\} \pmod{121}$—require 4 elements of H_{11}.

Proof. For any $y = a + 11b$, $(a + 11b)^{11} \equiv a^{11} \pmod{121}$. Since $3^5 = 243 = 2 \cdot 121 + 1 \equiv 1 \pmod{121}$, $(3^j)^{11} = 3^j(3^5)^{2j} \equiv 3^j \pmod{121}$, and since $\{\pm 3^j \bmod 11 : 0 \leq j \leq 4\} = (\mathbb{Z}/11\mathbb{Z})^\times$, the 10 non-zero signed 11-th powers modulo 121 are precisely H_{11} in (23). Because $\sum_{j=0}^{4} 3^j = 121$, every integer $u \in [-60, 60]$ has a unique balanced ternary expansion $u = \sum_{j=0}^{4} c_j 3^j$ with $c_j \in \{-1, 0, +1\}$. The $\sum_{w=0}^{3} \binom{5}{w} 2^w = 131$ digit strings of Hamming weight $w \leq 3$ cover 111 distinct residues modulo 121, while the 10 remaining residues $\mathcal{E}_{121} = \{\pm 20, \pm 47, \pm 56, \pm 59, \pm 60\} \pmod{121}$ have representations of weight 4 in H_{11} (e.g., $20 = 27 - 9 + 3 - 1$, $47 \equiv -81 + 9 - 3 + 1$, $56 = 81 - 27 + 3 - 1$, $59 \equiv -81 + 27 - 9 + 1$, $60 = 81 - 27 + 9 - 3 \pmod{121}$). Since every element of H_{11} is a unit modulo 11, Lemma 2.5(iv) with $11^1 \parallel 11$ lifts $\Delta(11, 11^2) = 4$ to $\Delta(11, 11^l) = 4$ for all $l \geq 2$. □

Theorem 4.4 (Sharper Upper Bound $v(11) \leq 28$)**.** *Every integer $n \in \mathbb{Z}$ can be expressed as an algebraic sum of at most 28 eleventh powers of non-negative integers:*

$$v(11) \leq 28. \tag{24}$$

Proof. By Proposition 4.2, the disjoint sets $A_+ = \{1, 7, 17, 30, 31, 36\}$ and $B_+ = \{3, 4, 19, 27, 34, 35\}$ satisfy $\sum_{a \in A_+} a^{2r} = \sum_{b \in B_+} b^{2r}$ for $2r = 2, 4, 6, 8$ (with common sums 3,496; 3,499,060; 3,817,541,236; and 4,337,082,467,140), while for $2r = 10$,

$$\sum_{a \in A_+} a^{10} - \sum_{b \in B_+} b^{10} = 5,068,293,003,419,476 - 5,034,947,307,035,476 = 33,345,696,384,000.$$

By Lemma 2.1(i) and (2), $A = \pm A_+$ and $B = \pm B_+$ have size $s = 12$ and degree 9, yielding the 24-term Frolov identity $\sum_{a \in A_+}[(x + a)^{11} + (x - a)^{11}] - \sum_{b \in B_+}[(x + b)^{11} + (x - b)^{11}] = C_{11}x$ with

$$C_{11} = 22 \sum_{i=1}^{6} (a_i^{10} - b_i^{10}) = 733,605,320,448,000 = 2^{11} \cdot 3^7 \cdot 5^3 \cdot 7^2 \cdot 11^2 \cdot 13 \cdot 17. \tag{25}$$

Evaluating $\Delta(11, p^l)$ across each prime power $p^l \parallel C_{11}$ via Lemma 2.5 and Lemma 4.3:

- For $p^l = 2^{11}$: Lemma 2.5(iii) gives $\Delta(11, 2^{11}) = 2$.

- For $p \in \{3, 5, 7, 13, 17\}$: Since $p - 1 \in \{2, 4, 6, 12, 16\}$ is coprime to 11, $\gcd(11, p-1) = 1$, so Lemma 2.5(i) gives $\Delta(11, 3^7) = \Delta(11, 5^3) = \Delta(11, 7^2) = 2$ and $\Delta(11, 13) = \Delta(11, 17) = 1$.
- For $p^l = 11^2$: Lemma 4.3 gives $\Delta(11, 11^2) = 4$.

Since $k = 11$ is odd, Theorem 2.3(i) gives $\Delta(11, C_{11}) = \max\{2, 2, 2, 2, 4, 1, 1\} = 4$, and (3) yields $v(11) \leq 2(12) + 4 = 28$. □

Remark 4.5 (Classification of Wróblewski's 71 Solutions)**.** In Wróblewski's catalogue [27], there are 71 coded degree-9, size-12 even multigrades (8.12.*). Computationally, 60 of these solutions satisfy $23 \mid C_{11}$ (yielding $v(11) \leq 35$), while all 11 solutions with $23 \nmid C_{11}$ (including 8.12.36) have no prime factor $p \equiv 1 \pmod{11}$ and independently establish $\Delta(11, C_{11}) = 4$ and $v(11) \leq 28$. The theorem $v(11) \leq 28$ is fully and unconditionally proved by the explicit single example 8.12.36 above.

4.3 Parity-Sign Selection ($v(11) \leq 27$) and Translation-Invariance Analysis

In Theorem 4.4, the symmetric solution 8.12.36 had constant term $D = S_{11}(A, B) = 0$. For a non-symmetric degree-9, size-12 PTE solution, choosing the sign $\epsilon \in \{+1, -1\}$ in $\epsilon P(x) = \epsilon(Cx + D)$ could reduce the 11-adic step from 4 to 3.

Theorem 4.6 (Conditional Parity-Sign Reduction to $v(11) \leq 27$ and Translation Invariance)**.** *Suppose $A = \{a_1, \ldots, a_{12}\}$ and $B = \{b_1, \ldots, b_{12}\}$ form a PTE solution of size $s = 12$ and degree 9 such that $C = 11S_{10}(A, B)$ and $D = S_{11}(A, B)$ satisfy:*

(a) $11^2 \mid C$ *and* $\gcd(11, p - 1) = 1$ *for every prime divisor $p \neq 11$ of C; and*

(b) $11^2 \nmid D = \sum_{i=1}^{12} a_i^{11} - \sum_{i=1}^{12} b_i^{11}$.

Then $v(11) \leq 24 + 3 = 27$. Moreover, for any degree-9, size-12 solution $A \overset{9}{=} B$, Newton's identities yield

$$D = \frac{e_1(A)}{11} C + 11\big(e_{11}(A) - e_{11}(B)\big), \tag{26}$$

so under $11^2 \mid C$, the condition $11^2 \nmid D$ is equivalent to

$$\frac{C}{121} e_1(A) + e_{11}(A) - e_{11}(B) \not\equiv 0 \pmod{11}, \tag{27}$$

and is invariant under affine translation $(A, B) \mapsto (A + t, B + t)$ $(t \in \mathbb{Z})$.

Proof. By Fermat's Little Theorem and $A \overset{9}{=} B$, $D = \sum_{i=1}^{12} a_i^{11} - \sum_{i=1}^{12} b_i^{11} \equiv S_1(A, B) = 0 \pmod{11}$. Combined with $121 \nmid D$, we have $D \equiv 11u \pmod{121}$ for some unit $u \in (\mathbb{Z}/11\mathbb{Z})^\times$, so $2D \equiv 22u \not\equiv 0 \pmod{121}$. By Lemma 4.3, the 10 extremal classes modulo 121 requiring 4 signed 11-th powers are $\mathcal{E}_{121} = \{\pm 20, \pm 47, \pm 56, \pm 59, \pm 60\}$ mod 121. Modulo 11, these 10 classes reduce to

$$56 \equiv 1, \quad -20 \equiv 2, \quad 47 \equiv 3, \quad 59 \equiv 4, \quad 60 \equiv 5 \pmod{11} \quad \text{and their negations},$$

mapping bijectively onto the 10 non-zero residue classes in $(\mathbb{Z}/11\mathbb{Z})^\times$. Thus each congruence class modulo 11 contains at most one element of $\mathcal{E}_{121}$. For any $n \in \mathbb{Z}$, $n - D$ mod 121 and $n + D$ mod 121 lie in the same congruence class modulo 11 (since $11 \mid D$) and are distinct modulo 121 (since $2D \not\equiv 0 \pmod{121}$), so they cannot both belong to $\mathcal{E}_{121}$. Hence there exists $\epsilon \in \{+1, -1\}$ such that $n - \epsilon D$ mod $121 \notin \mathcal{E}_{121}$, which is a sum of at most 3 signed 11-th powers modulo 121 (and modulo 11^l by Lemma 2.5(iv)). Since $\Delta(11, p^l) \leq 2 \leq 3$ for all

$p \neq 11$ dividing C by hypothesis (a), Theorem 2.3(i) gives $n - \epsilon D \equiv \sum_{j=1}^{3} \epsilon_j y_j^{11} \pmod{|C|}$, so $n = \epsilon(Cx + D) + \sum_{j=1}^{3} \epsilon_j y_j^{11}$ uses at most $24 + 3 = 27$ signed eleventh powers.

Finally, subtracting Newton's identities at $r = 10$ and $r = 11$ for $A \overset{9}{=} B$ gives $S_{10} = -10(e_{10}(A) - e_{10}(B))$ and $S_{11} - e_1(A)S_{10} = 11(e_{11}(A) - e_{11}(B))$. Substituting $S_{10} = C/11$ yields (26). Under $121 \mid C$, $D = 11[(C/121)e_1(A) + e_{11}(A) - e_{11}(B)]$, so $121 \nmid D \iff (C/121)e_1(A) + e_{11}(A) - e_{11}(B) \not\equiv 0 \pmod{11}$, proving (27). Translating every entry of A and B by $t \in \mathbb{Z}$ changes $D = S_{11}$ by tC; since $121 \mid C$, $tC \equiv 0 \pmod{121}$, so D mod 121 is invariant under translation. The upper bound $v(11) \leq 27$ remains conditional on the existence of a solution satisfying hypotheses (a) and (b). □

5 Modulo-29 Obstruction Bypass for $k = 14$ and Proof of $v(14) \leq 45$

We next analyze $k = 14$, where Frolov's method requires a degree-12 PTE solution. Since no ideal degree-12 solution ($s = 13$) is known, the smallest known degree-12 PTE solutions have size $s = 20$ ($2s = 40$) and are obtained in Wróblewski's catalogue [27, Pages 44–54] by applying Tarry's shift theorem to degree-11 solutions.

Proposition 5.1 (Tarry's Shift Theorem and the $p = 29$ Obstruction at $k = 14$)**.** *Let (A_0, B_0) be a PTE solution of size s_0 and degree m, and let $h \in \mathbb{Z} \setminus \{0\}$. Then the shifted multisets*

$$A_{\text{raw}} = A_0 \cup (B_0 + h), \quad B_{\text{raw}} = B_0 \cup (A_0 + h)$$

satisfy $A_{\text{raw}} \overset{m+1}{=} B_{\text{raw}}$ with $(m+2)$-nd power-sum difference

$$S_{m+2}(A_{\text{raw}}, B_{\text{raw}}) = -(m+2)hS_{m+1}(A_0, B_0). \tag{28}$$

Canceling the $c = |A_{\text{raw}} \cap B_{\text{raw}}|$ common elements from both sides yields a disjoint degree-$(m+1)$ PTE solution (A, B) of reduced size $s = 2s_0 - c$, and subsequent affine reduction by shift $t \in \mathbb{Z}$ and scale divisor $g \geq 1$ yields $S_{m+2}(\tilde{A}, \tilde{B}) = -(m+2)hg^{-(m+2)}S_{m+1}(A_0, B_0)$.

In particular, for every symmetric ideal degree-11 PTE solution (A_0, B_0) of size $s_0 = 12$ over $\mathbb{Z}$, Coppersmith et al. [9, Table 1] proved that $29 \mid C'_{12}(\mathbb{Z})$, so $29 \mid S_{12}(A_0, B_0) = -12C_{12}(A_0, B_0)$. Whenever $h = a_i - a_j > 0$ is chosen as a repeated difference within A_0 or B_0 ($c = 4$, giving size $s = 24 - 4 = 20$), provided $29 \nmid g$ for the resulting scaled solution, the resulting degree-12 solution $(\tilde{A}, \tilde{B})$ has $29 \mid C_{14} = 14S_{13}(\tilde{A}, \tilde{B})$, forcing $\Delta(14, |C_{14}|) \geq \Delta(14, 29) = 14$ and restricting the global bound to $2s + \Delta(14, |C_{14}|) \geq 40 + 14 = 54$.

Proof. For any $r \geq 1$, binomial expansion gives $S_r(A_{\text{raw}}, B_{\text{raw}}) = -\sum_{j=1}^{r-1} \binom{r}{j} h^{r-j} S_j(A_0, B_0)$. Since $S_j(A_0, B_0) = 0$ for $1 \leq j \leq m$, we obtain $S_r(A_{\text{raw}}, B_{\text{raw}}) = 0$ for $1 \leq r \leq m+1$, while at $r = m+2$ only the $j = m+1$ term survives, giving (28). Removing common elements preserves power-sum differences, and scaling divides S_{m+2} by g^{m+2}. For $m = 11$ ($s_0 = 12$), $29 \mid C'_{12}(\mathbb{Z})$ implies $29 \mid S_{12}(A_0, B_0)$, so whenever $29 \nmid g$, $29 \mid C_{14}$. Since $29 = 2 \cdot 14 + 1$ is prime, Lemma 2.5(ii) gives $\Delta(14, 29) = 14$. Indeed, all eight ideal-derived size-20 solutions in Wróblewski [27, Pages 46–54] satisfy $29 \mid C_{14}$ and give $v(14) \leq 54$. □

Theorem 5.2 (Sharper Upper Bound $v(14) \leq 45$ via Modulo-29 Bypass and Exact Signed-CRT Analysis)**.** *Every integer $n \in \mathbb{Z}$ can be expressed as an algebraic sum of at most 45 fourteenth powers of non-negative integers:*

$$v(14) \leq 45. \tag{29}$$

Proof. Instead of shifting an ideal size-12 degree-11 solution (which forces $29 \mid C_{14}$), we use Wróblewski's odd-power multigrade 11.20.107 [27, Page 44], obtained by Tarry-shifting the non-ideal size-16, degree-11 symmetric solution 10.16.93 (where $A_{0,+} = \{1, 29, 41, 44, 52, 72, 87, 93\}$ and $B_{0,+} = \{8, 33, 36, 39, 61, 67, 89, 92\}$) by $h = 28$. Because the difference $h = 28$ occurs 6 times among positive elements of 10.16.93 ($29 - 1 = 72 - 44 = 28$ in $A_{0,+}$ and $36 - 8 = 61 - 33 = 67 - 39 = 89 - 61 = 28$ in $B_{0,+}$), $c = 12$ elements cancel out of $2s_0 = 32$, leaving a degree-12 solution of size $s = 32 - 12 = 20$:

$$A_{\text{odd}} = \{13, 19, 25, 43, 55, 66, 78, 86, 101, 107\}, \quad B_{\text{odd}} = \{6, 27, 30, 38, 50, 73, 79, 81, 103, 106\}.$$

Direct computation verifies that $\sum_{a \in A_{\text{odd}}} a^{2r-1} = \sum_{b \in B_{\text{odd}}} b^{2r-1}$ for all $2r - 1 \in \{1, 3, 5, 7, 9, 11\}$ with exact common sums:

$$\begin{aligned}
&2r - 1 = 1 : 593, && 2r - 1 = 3 : 3,924,011,\\
&2r - 1 = 5 : 34,042,300,763, && 2r - 1 = 7 : 327,406,508,617,571,\\
&2r - 1 = 9 : 3,325,215,610,528,647,563, && 2r - 1 = 11 : 34,876,960,347,189,348,946,931.
\end{aligned}$$

By Lemma 2.1(ii), $A = A_{\text{odd}} \cup (-B_{\text{odd}})$ and $B = -A$ satisfy $A \overset{12}{=} B$ of size $s = 20$, yielding the 40-term Frolov identity $P_{14}(x) = \sum_{a \in A}(x + a)^{14} - \sum_{b \in B}(x + b)^{14} = C_{14}x$ (with $D_{14} = 0$) where

$$\begin{aligned}
C_{14} &= 28 \sum_{i=1}^{10} (a_i^{13} - b_i^{13}) = 10,092,027,867,805,880,156,160,000 \\
&= 2^{22} \cdot 3^8 \cdot 5^4 \cdot 7^4 \cdot 11^2 \cdot 13^2 \cdot 17 \cdot 19 \cdot 37.
\end{aligned} \tag{30}$$

Notice that $29 \nmid C_{14}$. We first evaluate the prime-power constants $\Delta(14, p^l)$ across $p^l \parallel C_{14}$:

- $p^l = 2^{22}$: Since $2^1 \parallel 14$, Lemma 2.5(iii) shows that the odd 14-th powers stabilize at $2^{1+2} = 8$ as $\{u \in (\mathbb{Z}/2^{22}\mathbb{Z})^\times : u \equiv 1 \pmod 8\}$, while even bases satisfy $2^{14} \mid y^{14}$. Thus every residue modulo 2^{22} is a sum of at most 4 signed odd 14-th powers (with $u \equiv 4 \pmod 8$ requiring 4 odd powers, whereas $3^{14} - 5^{14} \equiv 25 - 9 = 16 \pmod{32}$ requires only 2), giving $\Delta(14, 2^{22}) = 4$.
- $p^l = 7^4$: Since $7^1 \parallel 14$ and $\gcd(14, \phi(49)) = \gcd(14, 42) = 14$, the unit 14-th powers modulo 49 form the subgroup $\{1, 18, 30\}$ mod 49 of order 3, so the signed 14-th powers modulo 49 are $\{0, \pm 1, \pm 18, \pm 19\}$ mod 49. Direct verification shows that every residue modulo 49 is a sum of at most 4 elements of $\{\pm 1, \pm 18, \pm 19\}$ mod 49 with at least one unit summand, so Lemma 2.5(iv) gives $\Delta(14, 7^4) = \Delta(14, 49) = 4$.
- $p \in \{3, 11, 19\}$: Here $p \nmid 14$, $\gcd(14, p - 1) = 2$, and $p \equiv 3 \pmod 4$, so $-1 \notin (\mathbb{F}_p^\times)^2$ and the signed 14-th powers $\pm y^{14}$ cover all of $\mathbb{F}_p$, giving $\Delta(14, p) = 1$. By Lemma 2.5(i), $\Delta(14, 3^8) = \Delta(14, 11^2) = 2$ and $\Delta(14, 19) = 1$.
- $p \in \{5, 13, 17, 37\}$: Here $p \nmid 14$, $\gcd(14, p - 1) = 2$, and $p \equiv 1 \pmod 4$, so $\pm y^{14} \bmod p = (\mathbb{F}_p^\times)^2 \cup \{0\}$. Since every element of $\mathbb{F}_p$ is a sum of two squares, $\Delta(14, p) = 2$, and Lemma 2.5(ii) gives $\Delta(14, 5^4) = \Delta(14, 13^2) = \Delta(14, 17) = \Delta(14, 37) = 2$.

Although $\max_{p^l \parallel C_{14}} \Delta(14, p^l) = 4$, $k = 14$ is even, so Theorem 2.3(ii) applies.

Local Obstruction at $r = 4$: For any residue u satisfying $u \equiv 2 \pmod 8$ and $u \equiv 26 \pmod{49}$ (such as $u \equiv 26 \pmod{392}$):

- Modulo 2^{22}: since $y^{14} \in \{0, 1\} \pmod 8$, reaching $+2 \pmod 8$ with 4 terms requires at least two positive signs, so $S_4(14, 2^{22}; u) \subseteq \{2, 3, 4\}$.

- Modulo 49: in $\{0,1,18,30\} \bmod 49$, $26 \equiv -(18+18+18+18) \equiv 18-(30+30+30) \pmod{49}$ can only be reached in 4 terms with $P \in \{0,1\}$ positive signs, so $S_4(14,49;26) = \{0,1\}$.

Because $\{2,3,4\} \cap \{0,1\} = \emptyset$, the admissible sign-split sets are disjoint at $r = 4$, proving $\Delta(14, C_{14}) > 4$.

Global Resolution at $r = 5$: For all seven prime powers $p^l \in \{3^8, 5^4, 11^2, 13^2, 17, 19, 37\}$, we have $\Delta(14, p^l) \leq 2$. At $r = 5 \geq 2+2$, padding with $\pm 0^{14}$ ensures that every residue class modulo p^l has an admissible sign-split set containing all $P \in \{0,1,2,3,4,5\}$ (bitmask `0b111111`). Thus the 9-way intersection $\bigcap_{j=1}^{9} S_5(14, p_j^{l_j}; u_j)$ reduces entirely to the intersection $S_5(14,8;u_8) \cap S_5(14,49;u_{49})$. Exhaustive verification over all $8 \times 49 = 392$ pairs $(u_8, u_{49}) \in \mathbb{Z}/8\mathbb{Z} \times \mathbb{Z}/49\mathbb{Z}$ confirms that every pair has non-empty intersection (overlapping in $\{2,3\}$; see Section 7). Hence $\Delta(14, C_{14}) = 5$, and (3) establishes $v(14) \leq 40 + 5 = 45$. □

6 Unified Analysis and Corrected Proofs of Frolov Bounds for $9 \leq k \leq 17$

As established in Section 1.1, the bounds $v(8) \leq 28$ [21, 14, 31] and $v(13) \leq 27$ [7] are already proved in the literature. We now provide complete, self-contained, and corrected proofs of the sharpest explicit Frolov bounds for the remaining exponents $k \in \{9, 10, 12, 15, 17\}$ (combining with $k = 5, 11, 14$ in Table 1).

Theorem 6.1 (Explicit Frolov Bounds for $k \in \{9,10,12,15,17\}$)**.** *Wright's modular Frolov method* (3) *combined with Theorem 2.3 and Lemma 2.5 establishes the following upper bounds on $v(k)$:*

$$v(9) \leq 29, \quad v(10) \leq 30, \quad v(12) \leq 33, \quad v(15) \leq 57, \quad v(17) \leq 64.$$

Proof. We verify each exponent in turn:

1. $k = 9$ ($v(9) \leq 29$): The ideal degree-7 symmetric PTE solution of size $s = 8$ [2], $A_+ = \{2, 16, 21, 25\}$ and $B_+ = \{5, 14, 23, 24\}$, gives a 16-term Frolov identity with

$$C_9 = 18 \sum_{i=1}^{4} (a_i^8 - b_i^8) = 87,178,291,200 = 2^{11} \cdot 3^5 \cdot 5^2 \cdot 7^2 \cdot 11 \cdot 13.$$

For $p^l = 3^5$, since $3^2 \parallel 9$ and $\phi(27) = 18$, every unit y coprime to 3 satisfies $y^9 \equiv \pm 1 \pmod{27}$, so r signed 9-th powers modulo 27 only cover $[-r, r] \bmod 27$ unless $r \geq 13$; by Lemma 2.5(iv), $\Delta(9, 3^5) = \Delta(9, 27) = 13$. For the remaining prime powers: $\Delta(9, 7^2) = \Delta(9, 7) = 3$ (since $\gcd(9,6) = 3 \implies y^9 \in \{0, \pm 1\} \bmod 7$), $\Delta(9, 2^{11}) = \Delta(9, 5^2) = \Delta(9, 13) = 2$, and $\Delta(9, 11) = 1$. Since $k = 9$ is odd, Theorem 2.3(i) gives $\Delta(9, C_9) = 13$ and $v(9) \leq 16 + 13 = 29$.

2. $k = 10$ ($v(10) \leq 30$): Letac's ideal degree-8 PTE solution of size $s = 9$ [2, 9], normalized to $A = \{0, 24, 30, 83, 86, 133, 157, 181, 197\}$ and $B = \{1, 17, 41, 65, 112, 115, 168, 174, 198\}$, gives an 18-term Frolov identity with

$$|C_{10}| = 10|S_9(A, B)| = 303,968,253,004,416,000 = 2^{10} \cdot 3^6 \cdot 5^3 \cdot 7^2 \cdot 11 \cdot 13 \cdot 17 \cdot 23 \cdot 29 \cdot 41.$$

For $p^l = 5^3 = 125$, since $5^1 \parallel 10$ and $\phi(25) = 20$, $y^{10} \equiv 0, \pm 1 \pmod{25}$ for all $y \in \mathbb{Z}$, forcing $\Delta(10, 5^3) = \Delta(10, 25) = 12$ by Lemma 2.5(iv). Crucially, because $-1 \equiv 7^{10} \pmod{25}$ is itself a

10th power, both signs ± 1 are 10th powers, so every residue modulo 25 admits all sign-splits $P \in \{0, 1, \ldots, 12\}$. For the other prime powers: $\Delta(10, 11) = 5$, $\Delta(10, 2^{10}) = 4$, $\Delta(10, 41) = 4$, $\Delta(10, 3^6) = \Delta(10, 7^2) = \Delta(10, 13) = \Delta(10, 17) = \Delta(10, 29) = 2$, and $\Delta(10, 23) = 1$. At $r = 12$, every prime power factor p^l has $\Delta(10, p^l) \leq 12$, and padding with 0s yields admissible sign-split masks containing all $P \in \{0, \ldots, 12\}$ (bitmask `0b1111111111111`). Thus $\bigcap_{j=1}^{10} S_{12}(10, p_j^{l_j}; u_j) \neq \emptyset$ for all residue tuples (see Section 7), proving $\Delta(10, |C_{10}|) = 12$ and $v(10) \leq 18 + 12 = 30$.

3. $k = 12$ **(**$v(12) \leq 33$**):** Chen Shuwen's odd-power multigrade 9.12.323 [27, Page 20], $A_{\text{odd}} = \{7, 91, 173, 269, 289, 323\}$ and $B_{\text{odd}} = \{29, 59, 193, 247, 311, 313\}$, reconstructs via Lemma 2.1(ii) with shift $t = 323$ and scale $g = 2$ to a size-12, degree-10 PTE solution giving a 24-term Frolov identity with

$$C_{12} = 291,227,584,995,099,777,024,000 = 2^{14} \cdot 3^9 \cdot 5^3 \cdot 7^2 \cdot 11^3 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 31 \cdot 37.$$

Locally: $\Delta(12, 2^{14}) = 8$ (since $2^2 \parallel 12 \implies$ odd $y^{12} \equiv 1 \pmod{16}$), $\Delta(12, 13) = 6$ (since $y^{12} \in \{0, 1\} \bmod 13$), $\Delta(12, 3^9) = \Delta(12, 27) = 4$, and $\Delta(12, p^l) \leq 3$ for all other factors. Because $k = 12$ is even, we apply Theorem 2.3(ii):

- At $r = 8$: for $u \equiv 7 \pmod{16}$ and $u \equiv 11 \equiv -2 \pmod{13}$, we have $S_8(12, 16; 7) = \{7, 8\}$ and $S_8(12, 13; 11) = \{0, 1, 2, 3, 4, 5, 6\}$, which are disjoint! Thus $\Delta(12, C_{12}) > 8$.
- At $r = 9$: all other 9 prime powers have $\Delta(12, p^l) \leq 4$, so at $r = 9 \geq 4 + 2$ their masks contain all $P \in \{0, \ldots, 9\}$. Exhaustive intersection across all $16 \times 13 = 208$ pairs $(u_{16}, u_{13}) \in \mathbb{Z}/16\mathbb{Z} \times \mathbb{Z}/13\mathbb{Z}$ confirms that $S_9(12, 16; u_{16}) \cap S_9(12, 13; u_{13}) \neq \emptyset$ for every pair (see Section 7). Hence $\Delta(12, C_{12}) = 9$ and $v(12) \leq 24 + 9 = 33$.

4. $k = 15$ **(**$v(15) \leq 57$**):**

- $\Delta(15, 11^2) = \Delta(15, 11) = 5$ (since $\gcd(15, 10) = 5 \implies y^{15} \in \{0, \pm 1\} \bmod 11$);
- $\Delta(15, 3^{10}) = \Delta(15, 9) = 4$;
- $\Delta(15, 5^8) = \Delta(15, 25) = 3$ and $\Delta(15, 7^3) = \Delta(15, 7) = 3$;
- $\Delta(15, 2^{18}) = \Delta(15, 13^2) = \Delta(15, 19) = 2$, and $\Delta(15, 17) = 1$.

Since $k = 15$ is odd, Theorem 2.3(i) gives $\Delta(15, C_{15}) = 5$ and $v(15) \leq 52 + 5 = 57$.

5. $k = 17$ **(**$v(17) \leq 64$**):**

$$A_+ = \{3, 5, 47, 53, 57, 59, 61, 89, 103, 111, 111, 115, 139, 143, 153\},$$
$$B_+ = \{1, 23, 41, 43, 51, 65, 73, 93, 95, 99, 123, 123, 127, 149, 151\}.$$

Direct computation confirms that $\sum_{a \in A_+} a^{2r} = \sum_{b \in B_+} b^{2r}$ for all $2r \in \{2, 4, 6, 8, 10, 12, 14\}$ with exact common sums:

$$S_2 = 135,079, \qquad S_4 = 2,042,541,823,$$
$$S_6 = 36,497,094,204,439, \qquad S_8 = 708,317,587,613,471,983,$$
$$S_{10} = 14,437,309,834,982,675,179,399, \qquad S_{12} = 303,716,088,746,099,658,627,467,743,$$
$$S_{14} = 65,263,275,440,561,580,658,824,183,239,59.$$

The 16th-power difference is

$$\sum_{a\in A_+} a^{16} - \sum_{b\in B_+} b^{16} = 39,645,163,794,589,840,400,449,536,000,000,$$

yielding the 60-term Frolov identity with slope

$$C_{17} = 34\sum_{i=1}^{15}(a_i^{16} - b_i^{16}) \qquad = 2^{39}\cdot 3^{11}\cdot 5^6\cdot 7^3\cdot 11^2\cdot 13^2\cdot 17^2\cdot 19\cdot 23.$$

Every prime $p \neq 17$ dividing C_{17} satisfies $\gcd(17, p-1) = 1$ ($\Delta(17, p^l) \leq 2$). For $p^l = 17^2 = 289$, $3^{17} \equiv 224 \pmod{289}$ has multiplicative order 16 in $(\mathbb{Z}/289\mathbb{Z})^\times$ and generates all 16 non-zero 17-th powers modulo 289 (whereas $2^{17} \equiv 155 \pmod{289}$ has order 8 and does not generate). Additive breadth-first search over $\{0\} \cup \langle 224\rangle$ confirms $\Delta(17, 17^2) = 4$. By Theorem 2.3(i), $\Delta(17, C_{17}) = 4$ and $v(17) \leq 60 + 4 = 64$. □

7 Computational Verification and Reproducibility Certificate

To guarantee complete auditability, the even-k exhaustive signed-CRT calculations and catalogue multigrade verifications were executed via the following self-contained Python script (Python 3.10+):

```python
def get_sign_split_masks(k, m, r):
    powers = sorted(list({pow(x, k, m) for x in range(m)}))
    pos_sets, neg_sets = [{0}], [{0}]
    for _ in range(r):
        pos_sets.append({(a + p) % m for a in pos_sets[-1] for p in powers})
        neg_sets.append({(a - p) % m for a in neg_sets[-1] for p in powers})
    masks = [0] * m
    for P in range(r + 1):
        Q = r - P
        neg_list = list(neg_sets[Q])
        for a in pos_sets[P]:
            for b in neg_list:
                masks[(a + b) % m] |= (1 << P)
    return masks

# k = 14 verification at r = 5:
m8, m49 = get_sign_split_masks(14, 8, 5), get_sign_split_masks(14, 49, 5)
assert all(m8[u8] & m49[u49] for u8 in range(8) for u49 in range(49))

# k = 12 verification at r = 8 and r = 9:
m16_8, m13_8 = get_sign_split_masks(12, 16, 8), get_sign_split_masks(12, 13, 8)
assert any((m16_8[u16] & m13_8[u13]) == 0 for u16 in range(16) for u13 in range(13))
m16_9, m13_9 = get_sign_split_masks(12, 16, 9), get_sign_split_masks(12, 13, 9)
assert all(m16_9[u16] & m13_9[u13] for u16 in range(16) for u13 in range(13))

# k = 10 verification at r = 12:
m25, m11, m8 = get_sign_split_masks(10, 25, 12), get_sign_split_masks(10, 11, 12),
    get_sign_split_masks(10, 8, 12)
assert all(m25[u25] & m11[u11] & m8[u8] for u25 in range(25) for u11 in range(11) for u8
    in range(8))
```

Acknowledgments

This paper was written with the assistance of AI.

Table 1: Summary of Classical Analytic, Prior/Ideal PTE, and Sharper Polynomial Identity Bounds for $v(k)$ ($k = 5$ and $8 \le k \le 17$). Literature results for $k = 8$ [21, 14, 31] and $k = 13$ [7] are included for completeness alongside the new and corrected bounds proved in this paper.

Exp. k	**Analytic** $G(k)+1$	**Prior / Ideal PTE Bound**	**Identity Code / Source**	**Prime Factorization of Slope** $\lvert C\rvert$ **& Bottleneck Prime Power(s)**	**Global** $\Delta(k,\lvert C\rvert)$	**Sharper** $v(k) \le$
$k = 5$	18	10 ([12])	Weighted (15)	$2 \cdot 3 \cdot 5 \cdot 13 \cdot 9857$ (square-free, $p \not\equiv 1 \bmod 5$)	1	**9**
$k = 8$	43	30 (ideal PTE)	S_3 ([21, 14])	Vaserstein–Habsieger S_3 identity (2^5: $\Delta = 16$)	16	28 ([21])
$k = 9$	51	34 ([26])	Borwein_deg7	$2^{11} \cdot 3^5 \cdot 5^2 \cdot 7^2 \cdot 11 \cdot 13$ (3^5: $\Delta = 13$)	13	**29**
$k = 10$	60	38 ([26])	Letac_deg8	$2^{10} \cdot 3^6 \cdot 5^3 \cdot 7^2 \cdot 11 \cdot 13 \cdot 17 \cdot 23 \cdot 29 \cdot 41$ (5^3: $\Delta = 12$)	12	**30**
$k = 11$	68	31 (8.10.313)	8.12.36 ([27])	$2^{11} \cdot 3^7 \cdot 5^3 \cdot 7^2 \cdot 11^2 \cdot 13 \cdot 17$ (bypasses 23; 11^2: $\Delta = 4$)	4	**28**
$k = 12$	77	48 ([26])	9.12.323 ([27])	$2^{14} \cdot 3^9 \cdot 5^3 \cdot 7^2 \cdot 11^3 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 31 \cdot 37$ (2^{14}, 13: max $= 8$, signed-CRT $\Delta = 9$)	9	**33**
$k = 13$	85	50 ([26])	10.12.151 ([7])	$2^{14} \cdot 3^{10} \cdot 5^3 \cdot 7^2 \cdot 11^2 \cdot 13^3 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31$ (13^3: $\Delta = 3$)	3	27 ([7])
$k = 14$	93	54 (11.20.178)	11.20.107 ([27])	$2^{22} \cdot 3^8 \cdot 5^4 \cdot 7^4 \cdot 11^2 \cdot 13^2 \cdot 17 \cdot 19 \cdot 37$ (bypasses 29; $2^{22}, 7^4$: max $= 4$, signed-CRT $\Delta = 5$)	5	**45**
$k = 15$	101	82 ([26])	12.26.79 ([8])	$2^{18} \cdot 3^{10} \cdot 5^8 \cdot 7^3 \cdot 11^2 \cdot 13^2 \cdot 17 \cdot 19$ (11^2: $\Delta = 5$)	5	**57**
$k = 17$	> 105	94 ([26])	14.30.153 ([27])	$2^{39} \cdot 3^{11} \cdot 5^6 \cdot 7^3 \cdot 11^2 \cdot 13^2 \cdot 17^2 \cdot 19 \cdot 23$ (17^2: $\Delta = 4$)	4	**64**

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